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Thermodynamic assessment of the Co-Nd system

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Abstract

By means of CALPHAD (CALculation of PHAse Diagram) technique, the phase diagram and thermodynamic data of the Co-Nd system were critically assessed. The Co-Nd system contains five solution phases (liquid, face-centered cubic, body-centered cubic, hexagonal close-packed, double hexagonal close-packed) modeled with the Redlich-Kister equation and 10 intermetallic compounds $Co_{17}Nd_2$, Co_5Nd, $Co_{19}Nd_5$, Co_7Nd_2, Co_3Nd, Co_2Nd, Co_3Nd_2, Co_3Nd_4, Co_2Nd_5 and $CoNd_3$, which are all treated as stoichiometric compounds. A set of self-consistent thermodynamic parameters of the Co-Nd system was obtained.
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Keywords: Co-Nd system; Thermodynamic properties; CALPHAD technique

1. Introduction

The intermetallic compounds formed by rare earth elements and transition metals are of particular interest regarding their outstanding magnetic properties [1] and their reversible absorption of hydrogen gas at room temperature and nearly atmospheric pressure [2]. It is well-known that the phase diagrams and thermodynamic properties of the alloy system are of great value to alloy design and processing.

This work deals with the assessment of the thermodynamic description of the Co-Nd system by means of the CALPHAD technique. In this method, the thermodynamic properties of alloy systems are studied using thermodynamic models for the Gibbs energy of individual phases. The thermodynamic parameters involved in the models are optimized from the experimental thermodynamics and phase diagram information.

2. Thermodynamic models

2.1. Unary phases

The Gibbs energy function $G_i^{\phi}(T) = {}^0G_i^{\phi}(T) - H_i^{\mathrm{SER}}$ (298.15 K) for the element i (i = Co, Nd) in the phase ϕ (ϕ = liquid, body-centered cubic (bcc), face-centered cubic (fcc) or hexagonal close-packed (hcp), double hexagonal close-packed (dhcp)) is described by an equation of the following form:

$$G_i^{\phi}(T) = a + bT + cT\ \ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^9 \quad (1)$$

where H_i^{SER} (298.15 K) is the molar enthalpy of the element i at 298.15 K in its standard element reference (SER) state, hcp for Co and dhcp for Nd. The Gibbs energy of the element i, $G_i^{\phi}(T)$, in its SER state, is denoted by GHSER_i, i.e.,

$$\mathrm{GHSER_{Co}} = {}^0G_{\mathrm{Co}}^{\mathrm{hcp}}(T) - H_{\mathrm{Co}}^{\mathrm{SER}}\,(298.15\,\mathrm{K}) \quad (2)$$

$$\mathrm{GHSER_{Nd}} = {}^0G_{\mathrm{Nd}}^{\mathrm{dhcp}}(T) - H_{\mathrm{Nd}}^{\mathrm{SER}}\,(298.15\,\mathrm{K}) \quad (3)$$

In the present work, the Gibbs energy functions are taken from the SGTE compilation of Dinsdale [3].

2.2. Solution phases

In the Co-Nd system, there are four solution phases: liquid, bcc(Nd), fcc(Co) and hcp(Co), dhcp(Nd). Their Gibbs energies are described by the following expression:

$$G_{\mathrm{m}}^{\phi} = x_{\mathrm{co}}G_{\mathrm{Co}}^{\phi}(T) + x_{\mathrm{Nd}}G_{\mathrm{Nd}}^{\phi}(T) + RT(x_{\mathrm{Co}}\ \ln x_{\mathrm{Co}} + x_{\mathrm{Nd}}\ \ln x_{\mathrm{Nd}}) + {}^{\mathrm{E}}G_{\mathrm{m}}^{\phi} + {}^{\mathrm{mg}}G_{\mathrm{m}}^{\phi} \quad (4)$$

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where ${}^{E}G_{m}^{\phi}$ is the excess Gibbs energy, expressed by the Redlich-Kister polynomial [4],

$$ {}^{E}G_{m}^{\phi} = x_{Co}x_{Nd}\sum_{i} {}^{i}L^{\phi}(x_{Co} - x_{Nd})^{i} \quad (5) $$

where ${}^{i}L^{\phi}$ is the interaction parameter between elements Co and Nd, which is to be evaluated in the present work. Its general form is

$$ L^{\phi} = a + bT + cT\ \ln T + dT^{2} + eT^{3} + fT^{-1} \quad (6) $$

but there are cases when only the first one or two terms are used according to the temperature dependence on the experimental data. ${}^{mg}G_{m}^{\phi}$ is the magnetic contribution to the Gibbs energy, which is discussed in Section 2.4.

2.3. Stoichiometric intermetallic compounds

In the present work, the 10 intermetallic compounds $Co_{17}Nd_2$, Co_5Nd, $Co_{19}Nd_5$, Co_7Nd_2, Co_3Nd, Co_2Nd, Co_3Nd_2, Co_3Nd_4, Co_2Nd_5 and $CoNd_3$ are treated as stoichiometric compounds. The Gibbs energy per mole of formula unit Co_mNd_n is expressed as follows:

$$ G_{m}^{Co_mNd_n} = m\mathrm{GHSER}_{Co} + n\mathrm{GHSER}_{Nd} + \Delta G_{f}^{Co_mNd_n} + {}^{mg}G_{m}^{Co_mNd_n} \quad (7) $$

where $\Delta G_{f}^{Co_mNd_n}$ is the Gibbs energy of formation per mole of formula unit Co_mNd_n. Because of lack of experimental measurements, it is assumed that the Neumann-Kopp rule applies to the heat capacity, i.e. $\Delta C_{p_f} = 0$. Thus $\Delta G_{f}^{Co_mNd_n}$ can be given by the following expression:

$$ \Delta G_{f}^{Co_mNd_n} = a + bT \quad (8) $$

where the parameters a and b were evaluated in the present work. ${}^{mg}G_{m}^{Co_mNd_n}$ is the magnetic contribution to the Gibbs energy, which is discussed in Section 2.4.

2.4. The magnetic contribution to the Gibbs energy

The magnetic contribution to the Gibbs energy ${}^{mg}G_{m}$ is described by

$$ {}^{mg}G_{m} = RT\ \ln(\beta + 1)f(\tau) \quad (9) $$

where β is a quantity related to the total magnetic entropy, which in most cases is set equal to the Bohr magnetic moment per mole of atoms; τ is defined as T/T_c, and T_c is the critical temperature for magnetic ordering, i.e. the Curie temperature (T_c) for ferromagnetic ordering and the Néel temperature (T_N) for antiferromagnetic ordering; $f(\tau)$ represents the polynomials obtained by Hillert and Jarl [5] based on the magnetic specific heat of iron, i.e.

for $\tau < 1$:

$$ f(\tau) = 1 - \frac{79\tau^{-1}/(140p) + 474/497(1/p - 1)(\tau^{3}/6 + \tau^{9}/135 + \tau^{15}/600)}{A} \quad (10) $$

for $\tau > 1$:

$$ f(\tau) = -\frac{\tau^{-5}/10 + \tau^{-15}/315 + \tau^{-25}/1500}{A} \quad (11) $$

where $A = 519/1125 + 11692/15975(1/p - 1)$ and p depends on the structure, 0.4 for bcc structure and 0.28 for the others.

3. Experimental information

The structure of intermetallic compounds found in the Co-Nd binary system has been systematically studied by Singh and Raman [6]. Ray et al. [7–10] measured the complete phase diagram of the Co-Nd system, which included 10 intermetallic compounds: $Co_{17}Nd_2$, Co_5Nd, $Co_{19}Nd_5$, Co_7Nd_2, Co_3Nd, Co_2Nd, Co_3Nd_2, $Co_{1.7}Nd_2$, $Co_{\sim3}Nd_{\sim7}$ and $CoNd_3$. Later, Wu et al. [11] reinvestigated the Co-Nd phase diagram in the range from 0 to 60 at.% Nd and above 600 °C, and specified seven intermetallic compounds exist, $Co_{17}Nd_2$, Co_5Nd, $Co_{19}Nd_5$, Co_7Nd_2, Co_3Nd, Co_2Nd and Co_3Nd_2, in which the compound $Co_{17}Nd_2$ is formed congruently with a eutectic reaction liquid → $Co_{17}Nd_2$ + fcc(Co) at 92.9 at.% Co at 1370 °C, others are formed by peritectic reaction, and Co_5Nd decomposes into $Co_{17}Nd_2$ and $Co_{19}Nd_5$ at lower 800 °C. In the Co-rich isothermal section of the Co-Nd-Pr ternary system at 500 °C measured by Zhuang et al. [12], it confirmed the existence of Co_5Nd and Co_3Nd_4, which originally was considered as $Co_{1.7}Nd_2$ [10]. In the present work, Co_5Nd is treated as stable at room temperature. The solid solubility of Nd in fcc(Co) is extremely small, estimated to be 0.03 at.% at 1000 °C [11]. Based on the Wu et al. [11] (<60 at.% Nd) and Ray [10] (>50 at.% Nd) investigation, a combination of the Co-Nd phase diagram can be obtained, in which $Co_{\sim3}Nd_{\sim7}$ [10] is changed into Co_2Nd_5 based on the crystallographic data determined by Moreau and Paccard [13].

The enthalpies of formation of intermetallic compounds $Co_{17}Nd_2$, Co_5Nd, $Co_{19}Nd_5$, Co_7Nd_2, Co_3Nd, Co_2Nd, Co_3Nd_2, Co_3Nd_4, Co_2Nd_5 and $CoNd_3$ were predicted using revised Miedema model [14] in the present work, in kJ mol^{-1} of atoms: $Co_{17}Nd_2$, −10.8; Co_5Nd, −17.0; $Co_{19}Nd_5$, −20.9; Co_7Nd_2, −22.2; Co_3Nd, −24.3; Co_2Nd, −28.9; Co_3Nd_2, −30.2; Co_3Nd_4, −25; Co_2Nd_5, −17.1; $CoNd_3$, −14.9.

Buschow [15] compiled magnetic data of the intermetallic compounds of rare earth and transition metals. The values of the critical temperature for magnetic ordering and the Bohr magnetic moment of the compounds, which are used in the present work, are listed in Table 1.

4. Assessment procedure

In the Co-Nd system, there are five solution phases: liquid, bcc, fcc, hcp and dhcp. As the dhcp structure is not stable for pure Co and the hcp structure is not stable for pure Nd, their lattice stabilities are required. Therefore according to the estimation of Fernandez Guillermet and Huang [16] for high melting bcc metals V, Nb and Ta, the following lattice stabilities are estimated,

$$ {}^{0}G_{Co}^{dhcp} = {}^{0}G_{Co}^{hcp} + 5000\ \mathrm{J\ mol^{-1}} \quad (12) $$

$$ {}^{0}G_{Nd}^{hcp} = {}^{0}G_{Nd}^{dhcp} + 5000\ \mathrm{J\ mol^{-1}} \quad (13) $$

The solid solubility of Nd in fcc(Co) is estimated to be 0.03 at.% at 1000 °C [11]. In the present work, ${}^{0}L_{Co,Nd}^{bcc}$, ${}^{0}L_{Co,Nd}^{dhcp}$ and ${}^{0}L_{Co,Nd}^{fcc}$ are set to be equal to +2000 J mol^{-1}, i.e.

$$ {}^{0}L_{Co,Nd}^{bcc} = {}^{0}L_{Co,Nd}^{fcc} = {}^{0}L_{Co,Nd}^{dhcp} = +2000\ \mathrm{J\ mol^{-1}} \quad (14) $$

Table 1
Measured T_c and β (μ_B) values in the Co-Nd system[a]

Phase	$Co_{17}Nd_2$	Co_5Nd	$Co_{19}Nd_5$	Co_7Nd_2	Co_3Nd	Co_2Nd	$CoNd_3$[b]
T_c (K)	1157	910	714	609	392.6	105.3	14
β (μ_B)	33.5	10.6	31.0	13.1	5.52	3.45	4.5

[a] In SI units per mole of formula units.
[b] Antiferromagnetic ordering temperature T_N for $CoNd_3$.

The optimization was carried out by means of the Thermo-Calc software [17] that can handle various kinds of experimental data. The thermodynamic parameters were evaluated using the PARROT module in the Thermo-Calc software [17]. This module works by minimizing an error sum where each piece of information is given a certain weight according to its estimated accuracy and allows the simultaneous consideration of various types of thermodynamic and phase diagram data. The best-fit criterion follows the maximum likelihood principle and that principle as a criterion for best fit is the most efficient when the amount of data is large.

5. Results and discussion

5.1. The assessed phase diagram

The completed thermodynamic parameters of the Co-Nd system obtained in the present work are listed in Table 2. The invariant equilibria in the Co-Nd system are listed in Table 3, in which the reviewed data by Okamoto [18] is presented. The Co-Nd phase diagram calculated by the present thermodynamic parameters is shown in Fig. 1, and nearly identical to the one reported by Ray [10] and Wu et al. [11]. The main difference is the melting behavior of $Co_{17}Nd_2$. The microstructure of as-cast alloy containing 92.9 at.% Co shown by Wu et al. [11] is the typical dendritic structure ([11], p. 164, Fig. 4) as can be seen from Fig. 2, and does not exhibit literally the characteristics of typical eutectic structure. The similar events also occur in the microstructure of as-cast alloy containing 92.9 at.% Co in the Co-Pr system [19], 87.0, 92.9 at.% Co in the Co-Ce system [20], 42.86 at.% Co in the Co-Dy system [21] and 37.0 at.%

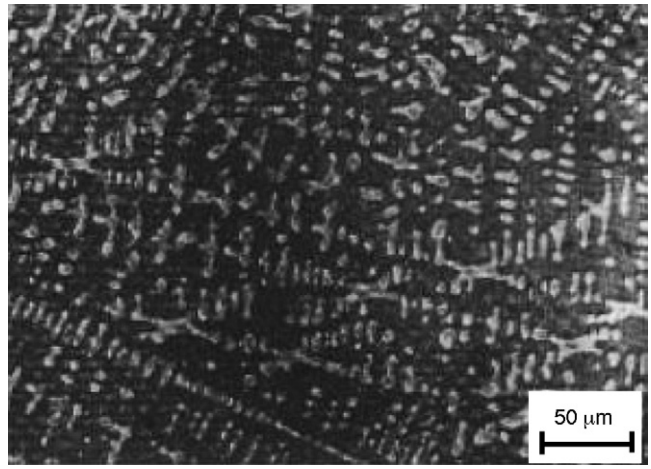

Fig. 2. Microstructure of 92.9 at.% Co alloy as-cast in the Co-Nd system reported by Wu et al. [11], which is a typical dendritic structure.

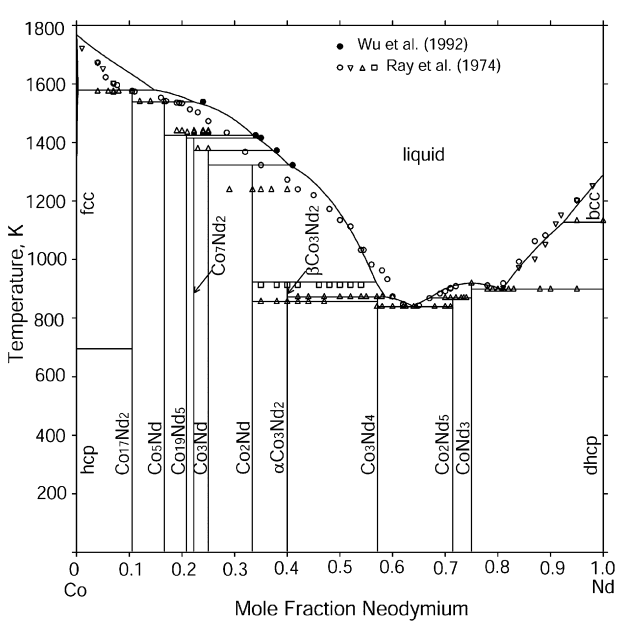

Fig. 1. Calculated phase diagram of the Co-Nd system with the experimental data measured by Ray et al. [10] and Wu et al. [11].

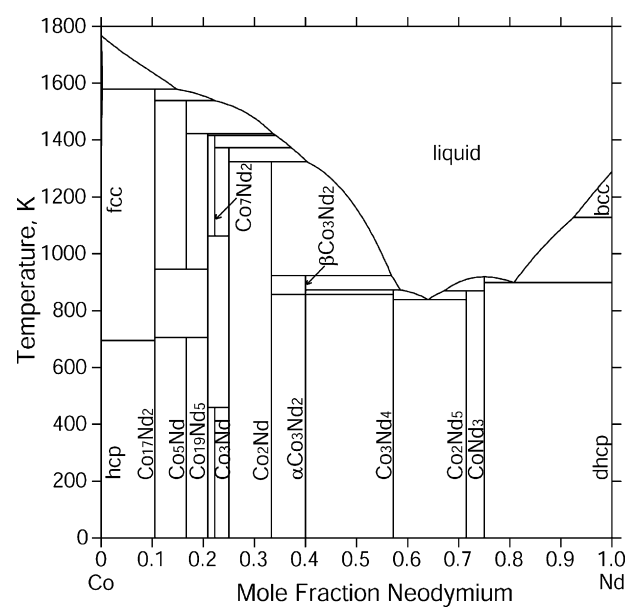

Fig. 3. Calculated phase diagram of the Co-Nd system, in which Co_5Nd and Co_7Nd_2 are unstable in the temperature range 705–945 and 459–1063 K, respectively.

Table 2
Optimized thermodynamic parameters of the Co-Nd system[a]

Co

$$\mathrm{GHSER_{Co}} = \begin{cases} +310.241 + 133.36601T - 25.0861T\ \ln T - 0.002654739T^2 - 1.7348 \times 10^{-7}T^{-3} + 72527T^{-1} & (298.14\,\mathrm{K} < T < 1768\,\mathrm{K}) \\ -17197.666 + 253.28374T - 40.5T\ \ln T + 9.3488 \times 10^{30}T^{-9} & (1768\,\mathrm{K} < T < 6000\,\mathrm{K}) \end{cases}$$

$$G^{\mathrm{liq.}} = \begin{cases} \mathrm{GHSER_{Co}} + 15085.037 - 8.931932T - 2.19801 \times 10^{-21}T^7 & (298.14\,\mathrm{K} < T < 1768\,\mathrm{K}) \\ -846.61 + 243.599944T - 40.5T\ \ln T & (1768\,\mathrm{K} < T < 6000\,\mathrm{K}) \end{cases}$$

$$G^{\mathrm{bcc}} = \mathrm{GHSER_{Co}} + 2938 - 0.7138T$$

$$T_{\mathrm{c}}^{\mathrm{bcc}} = 1450\,\mathrm{K}, \quad \beta^{\mathrm{bcc}}(\mu_{\mathrm{B}}) = 1.35$$

$$G^{\mathrm{fcc}} = \begin{cases} \mathrm{GHSER_{Co}} + 427.591 - 0.615248T & (298.14\,\mathrm{K} < T < 1768\,\mathrm{K}) \\ \mathrm{GHSER_{Co}} + 427.591 - 0.615253T & (1768\,\mathrm{K} < T < 6000\,\mathrm{K}) \end{cases}$$

$$T_{\mathrm{c}}^{\mathrm{fcc}} = 1396\,\mathrm{K}, \quad \beta^{\mathrm{fcc}}(\mu_{\mathrm{B}}) = 1.35$$

$$G^{\mathrm{hcp}} = \mathrm{GHSER_{Co}}$$

$$T_{\mathrm{c}}^{\mathrm{hcp}} = 1396\,\mathrm{K}, \quad \beta^{\mathrm{hcp}}(\mu_{\mathrm{B}}) = 1.35$$

Nd

$$\mathrm{GHSER_{Nd}} = \begin{cases} -8402.93 + 111.10239T - 27.0858T\ \ln T + 5.56125 \times 10^{-4}T^2 - 2.6923 \times 10^{-6}T^3 + 34887T^{-1} & (298.14\,\mathrm{K} < T < 900\,\mathrm{K}) \\ -6984.083 + 83.662617T - 22.7536T\ \ln T - 0.00420402T^2 - 1.802 \times 10^{-6}T^3 & (900\,\mathrm{K} < T < 1128\,\mathrm{K}) \\ -225610.846 + 1673.04075T - 238.182873T\ \ln T + 0.078615997T^2 - 6.048207 \times 10^{-6}T^3 + 38810350T^{-1} & (1128\,\mathrm{K} < T < 1799\,\mathrm{K}) \\ -25742.331 + 276.257088T - 48.7854T\ \ln T & (1799\,\mathrm{K} < T < 1800\,\mathrm{K}) \end{cases}$$

$$G^{\mathrm{liq.}} = \begin{cases} +5051.743 - 1.585076T + \mathrm{GHSER_{Nd}} & (298.14\,\mathrm{K} < T < 300\,\mathrm{K}) \\ +5350.01 - 86.593963T + 5.357301T\ \ln T - 0.046955463T^2 + 6.860782 \times 10^{-6}T^3 - 374380T^{-1} & (300\,\mathrm{K} < T < 1128\,\mathrm{K}) \\ -16335.232 + 268.625903T - 48.7854T\ \ln T & (1128\,\mathrm{K} < T < 1799\,\mathrm{K}) \\ +9407.099 - 7.631185T + \mathrm{GHSER_{Nd}} & (1799\,\mathrm{K} < T < 1800\,\mathrm{K}) \end{cases}$$

$$G^{\mathrm{bcc}} = \begin{cases} +1437.295 - 0.546281T + \mathrm{GHSER_{Nd}} & (298.14\,\mathrm{K} < T < 400\,\mathrm{K}) \\ +7312.2 - 153.033976T + 14.9956777T\ \ln T - 0.050479T^2 + 7.287217 \times 10^{-6}T^3 - 831810T^{-1} & (400\,\mathrm{K} < T < 1128\,\mathrm{K}) \\ -18030.266 + 239.677322T - 44.559T\ \ln T & (1128\,\mathrm{K} < T < 1289\,\mathrm{K}) \\ +334513.017 - 2363.9199T + 311.409193T\ \ln T - 0.156030778T^2 + 1.2408421 \times 10^{-5}T^3 - 64319604T^{-1} & (1289\,\mathrm{K} < T < 1800\,\mathrm{K}) \end{cases}$$

$$G^{\mathrm{fcc}} = \begin{cases} +500 + \mathrm{GHSER_{Nd}} & (298.14\,\mathrm{K} < T < 1128\,\mathrm{K}) \\ +500 + 9.99999884 \times 10^{-7}T + 3.00000011 \times 10^{-7}T\ \ln T + \mathrm{GHSER_{Nd}} & (1128\,\mathrm{K} < T < 1799\,\mathrm{K}) \\ +500 + \mathrm{GHSER_{Nd}} & (1799\,\mathrm{K} < T < 1800\,\mathrm{K}) \end{cases}$$

$$G^{\mathrm{dhcp}} = \mathrm{GHSER_{Nd}} \quad (298.14\,\mathrm{K} < T < 1800\,\mathrm{K})$$

Liquid

$${}^{0}L_{\mathrm{Co,Nd}}^{\mathrm{liq.}} = -92297 + 9.652T$$

$${}^{1}L_{\mathrm{Co,Nd}}^{\mathrm{liq.}} = -6060 + 20.427T$$

$${}^{2}L_{\mathrm{Co,Nd}}^{\mathrm{liq.}} = +71682 - 39.014T$$

bcc $\quad {}^{0}L_{\mathrm{Co,Nd}}^{\mathrm{bcc}} = +2000$

fcc $\quad {}^{0}L_{\mathrm{Co,Nd}}^{\mathrm{fcc}} = +2000$

dhcp $\quad {}^{0}L_{\mathrm{Co,Nd}}^{\mathrm{dhcp}} = +2000$

$\mathrm{Co_{17}Nd_2}$ $\quad \Delta G_{\mathrm{f}}^{\mathrm{Co_{17}Nd_2}} = -313664 + 94.459T$

$\mathrm{Co_5Nd}$ $\quad \Delta G_{\mathrm{f}}^{\mathrm{Co_5Nd}} = -125252 + 31.012T$

$\mathrm{Co_{19}Nd_5}$ $\quad \Delta G_{\mathrm{f}}^{\mathrm{Co_{19}Nd_5}} = -607724 + 219.708T - 7.707899T\ \ln T$

$\mathrm{Co_7Nd_2}$ $\quad \Delta G_{\mathrm{f}}^{\mathrm{Co_7Nd_2}} = -230419 + 58.917T$

$\mathrm{Co_3Nd}$ $\quad \Delta G_{\mathrm{f}}^{\mathrm{Co_3Nd}} = -106751 + 25.870T$

$\mathrm{Co_2Nd}$ $\quad \Delta G_{\mathrm{f}}^{\mathrm{Co_2Nd}} = -86543 + 17.408T$

$\alpha\mathrm{Co_3Nd_2}$ $\quad \Delta G_{\mathrm{f}}^{\alpha\mathrm{Co_3Nd_2}} = -147698 + 35.561T$

$\beta\mathrm{Co_3Nd_2}$ $\quad \Delta G_{\mathrm{f}}^{\beta\mathrm{Co_3Nd_2}} = -147228 + 35.012T$

$\mathrm{Co_3Nd_4}$ $\quad \Delta G_{\mathrm{f}}^{\mathrm{Co_3Nd_4}} = -187038 + 45.082T$

$\mathrm{Co_2Nd_5}$ $\quad \Delta G_{\mathrm{f}}^{\mathrm{Co_2Nd_5}} = -133113 + 13.951T$

$\mathrm{CoNd_3}$ $\quad \Delta G_{\mathrm{f}}^{\mathrm{CoNd_3}} = -67758 + 4.223T$

[a] In $\mathrm{J\,mol^{-1}}$ of the formula units.

Table 3
Invariant reactions in the Co-Nd system

Reaction	Present work				Okamoto [18]			
	T (K)		x(Nd)		T (K)		x(Nd)	
liq. + fcc(Co) → $Co_{17}Nd_2$[a]	1579	0.1477	0.0024	–	1579	0.15	0.00	–
fcc(Co) → $Co_{17}Nd_2$ + hcp(Co)	695	0.0000	–	0.0000	695	0.00	–	0.00
liq. + $Co_{17}Nd_2$ → Co_5Nd	1538	0.2227	–	–	1539	0.27	–	–
liq. + Co_5Nd → $Co_{19}Nd_5$	1425	0.3352	–	–	1425	0.35	–	–
liq. + $Co_{19}Nd_5$ → Co_7Nd_2	1416	0.3420	–	–	1416	0.36		–
liq. + Co_7Nd_2 → Co_3Nd	1373	0.3721	–	–	1373	0.37	–	–
liq. + Co_3Nd → Co_2Nd	1323	0.4035	–	–	1323	0.40	–	–
liq. + Co_2Nd → βCo_3Nd_2	923	0.5683	–	–	923	0.575	–	–
βCo_3Nd_2 → αCo_3Nd_2	856	–	–	–	856	–	–	–
liq. + βCo_3Nd_2 → Co_3Nd_4	872	0.5860	–	–	872	0.61	–	–
liq. → Co_3Nd_4 + Co_2Nd_5	839	0.6400	–	–	839	0.64	–	–
liq. + $CoNd_3$ → Co_2Nd_5	869	0.6712	–	–	869	0.69	–	–
liq. → $CoNd_3$	919	0.7500	–	–	919	0.75	–	–
liq. → $CoNd_3$ + dhcp(Nd)	898	0.8080	–	1.0000	898	0.80	–	1.00
liq. + bcc(Nd) → dhcp(Nd)	1128	0.9300	1.0000	1.0000	1128[b]	0.93	1.00	1.00

[a] The invariant reaction liq. → fcc(Co) + $Co_{17}Nd_2$ was reported by Wu et al. [11] and Okamoto [18].
[b] Based on SGTE pure element database compiled by Dinsdale [3].

Co in the Co-Sm system [22]. The microstructures of as-cast 90.6 at.% Co alloy in the Co-Sm system reported by Ge et al. [22] and 88.43 at.% Co alloy in the Co-Y system reported by Wu et al. [23] are the typical eutectic structure. In the present work, $Co_{17}Nd_2$ is treated as being formed by peritectic reaction liq. + fcc(Co) → $Co_{17}Nd_2$, and the peritectic reaction temperature is set to be equal to the melting temperature of $Co_{17}Nd_2$ at 1579 K. Based upon the above fact, Du and Lü [24] predicted that the compound $Co_{17}Y_2$ in the Co-Y system [23] was formed by peritectic reaction because the microstructure of 92.9 at.% Co alloy in the Co-Y system was not presented by Wu et al. [23]. However, Golumbfskie and Liu [25] treated $Co_{17}Y_2$ as a congruent melting compound based on the work of Wu et al. [23].

Because of the influence of the magnetic contribution to the Gibbs energy causing the instability of Co_5Nd in the temperature range between 705 and 945 K and Co_7Nd_2 in the temperature range between 459 and 1063 K as shown in Fig. 3, $\Delta G_f^{Co_{19}Nd_5}$ is assessed by the following expression:

$$\Delta G_f^{Co_{19}Nd_5} = a + bT + cT \ln T \quad (15)$$

5.2. Enthalpy of formation for intermetallic compounds

Fig. 4 shows the standard enthalpy of formation in the Co-Nd system and the comparison with the predicted values by the revised Miedema model [14] and the experimental data measured by Shilov [26] and Meyer-Liautad et al. [27] in the Co-Sm system. The enthalpies of formation for many compounds predicted by Miedema model [14] are usually more exothermic than the experimental results [28]. In addition, there was a steady increase in the exothermic character of the enthalpies of formation of the compounds formed by transition metals and rare earth elements with increasing number of f electrons of the rare earth elements [29,30]. Therefore, reasonable agreement should be obtained between the calculated results and the predicted values in the Co-Nd system.

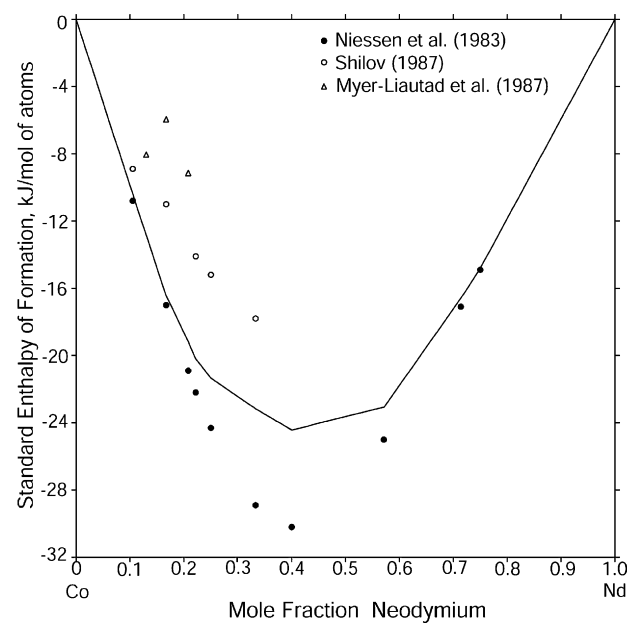

Fig. 4. Calculated enthalpies of formation at 298 K in the Co-Nd system in comparison with predicted values of the revised Miedema model [14] and the experimental data measured by Shilov [26] and Myer-Liautad et al. [27] in the Co-Sm system, respectively.

6. Conclusions

The phase relationships and thermodynamic description in the Co-Nd system were critically evaluated from the experimental information available in the literature. A self-consistent set of thermodynamic parameters was derived. With the thermodynamic description available, one can now make various calculations of practical interest.

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References

[1] K.H.J. Buschow, Rep. Prog. Phys. 40 (1977) 1179.
[2] K.H.J. Buschow, Mater. Res. Bull. 19 (1984) 935.
[3] A.T. Dinsdale, Calphad 15 (1991) 317.
[4] O. Redlich, A.T. Kister, Ind. Eng. Chem. 40 (1948) 345.
[5] M. Hillert, M. Jarl, Calphad 2 (1978) 227–238.
[6] P.P. Singh, A. Raman, Mater. Res. Bull. 3 (1968) 843.
[7] A.E. Ray, G.I. Hoffer, Proc. 8th Rare Earth Res. Conf., vol. 2, Reno, Nevada, 1970, p. 1524.
[8] A.E. Ray, A.T. Biermann, R.S. Harmer, J.E. Davison, Cobalt 4 (1973) 90.
[9] A.E. Ray, A.T. Biermann, R.S. Harmer, J.E. Davison, in: C.J. Kavane, T. Moeller (Eds.), Proc. 10th Rare Earth Res. Conf., vol. 2, Carfree, Arizona, 1973, p. 711.
[10] A.E. Ray, Cobalt 1 (1974) 13.
[11] C.H. Wu, Y.C. Chuang, X.M. Jin, X.H. Guan, Z. Metallkd. 83 (1992) 162.
[12] Y.H. Zhuang, J.L. Yan, H.Y. Zhou, X.L. Ou, J. Alloys Compd. 266 (1998) 164.
[13] J.M. Moreau, D. Paccard, Acta Crystallogr. B32 (1976) 1654.
[14] A.K. Niessen, F.R. de Boer, R. Boom, P. de Chatel, W.C.M. Mattens, A.R. Miedema, Calphad 7 (1983) 51.
[15] K.H.J. Buschow, Rep. Prog. Phys. 40 (1977) 1179.
[16] A. Fernandez Guillermet, W. Huang, Z. Metallkd. 79 (1988) 88.
[17] B. Sundman, B. Jansson, J.O. Andersson, Calphad 9 (1985) 153.
[18] H. Okamoto, J. Phase Equilib. 14 (1993).
[19] C.H. Wu, Y.C. Chuang, X.M. Jin, X.H. Guan, Z. Metallkd. 83 (1992) 32.
[20] C.H. Wu, Y.C. Chuang, X.M. Jin, X.H. Guan, Z. Metallkd. 82 (1991) 621.
[21] C.H. Wu, L.G. Yao, Y.C. Chuang, Z. Metallkd. 85 (1994) 104.
[22] W.Q. Ge, C.H. Wu, Y.C. Chuang, Z. Metallkd. 84 (1993) 165.
[23] C.H. Wu, Y.C. Chuang, X.P. Su, Z. Metallkd. 82 (1991) 73.
[24] Z. Du, D. Lü, J. Alloys Compd. 373 (2004) 171.
[25] W. Golumbfskie, Z.-K. Liu, J. Alloys Compd. 407 (2006) 193.
[26] A.L. Shilov, Zh. Fiz. Xhim. 61 (1987) 1384.
[27] F. Meyer-Liautad, C.H. Allibert, R. Castanet, J. Less-Common Met. 127 (1987) 243.
[28] Q. Guo, O.J. Kleppa, J. Alloys Compd. 321 (2001) 169.
[29] Q. Guo, O.J. Kleppa, Z. Metallkd. 87 (1996) 10.
[30] Q. Guo, O.J. Kleppa, J. Alloys Compd. 234 (1996) 280.